Zhexiang (Bruce) Li

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Internship availability: June 13 - September 17, 2027

Education

University of California, Los Angeles Sep 2026 - Jun 2028 (expected)
M.S. in Computer Science

University of Southern California May 2026
B.S. in Computer Science; GPA: 3.76/4.00; Viterbi Dean's List

Research Skills

Methods: multimodal learning, efficient attention, token routing, molecular property prediction
Tools: Python, PyTorch, Triton, RDKit, Chemprop, C/C++, SQL, Linux
Evaluation: patient-disjoint benchmarks, scaffold splits, hard negatives, ablation studies, SHAP

Research Experience

Graduate Researcher | Health Intelligence Lab, UCLA Aug 2026 - Present
Advisor: Prof. Yuzhe Yang

- Investigate multimodal foundation models for diabetes, combining CGM with insulin, meals, activity, and patient context for glucose forecasting.
- Built MetaboNet benchmarking and audit pipelines with patient-disjoint splits; compared GluFormer, T1DFormer, predictive-state models, and Transformers at 30/60/120-minute horizons.

Drug Discovery Research | USC, Prof. Yan Liu's group Spring 2026

- Worked with a Ph.D. student on antiviral drug discovery; compared RDKit, Chemprop, and hybrid models across seven Therapeutics Data Commons datasets using scaffold splits.
- Analyzed RDKit descriptor importance and SHAP attributions to examine which molecular features influenced predicted antiviral activity.

Selected Papers

MedTokenBudget: Lesion-Preserving Token Routing 2026
Zhexiang Li. Preprint. arXiv:2609.30613

- Developed a lesion-supervised router after the vision backbone, combining budget curriculum learning, diversity regularization, and attention distillation.
- Evaluated the fraction of annotated lesion evidence retained on ISIC 2019 under fixed token budgets.

Mass-Conservation Token Transformers for MS/MS Spectrum Retrieval 2026
Manuscript submitted to AAAI 2027

- Encoded mass-conserving fragment pairs as sparse tokens for spectrum retrieval; built closest-mass and overlap-hard negatives and tested ranking changes after removing evidence tokens.

Efficient Attention via Pre-Scoring 2025
First author; with H. Wang, Y. Bao, D. Woodruff. Under review. arXiv:2505.11040

- Proposed pre-scoring to select informative key intervals before attention; implemented the method in Triton.
- Benchmarked against HyperAttention and LevAttention to compare attention approximations.

Additional Experience

AI Engineer Intern | PMAOS / Next Big Seek Jul - Aug 2026

- Designed diversity-aware recommendation with session, bridge, and long-term exposure budgets, plus 11-language audio-visual affect inference with abstention on conflicting evidence.

Machine Learning Tutor | UCSC Spring 2023

- Led office hours on logistic regression, SVMs, and tree models; co-designed Jupyter assignments and contributed to the coding autograder for Prof. Lise Getoor's course.